

Sabrina Sousa

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Experience

Technical Consultant | Contax Inc. May 2023 - Aug 2024

- Resolved client needs with custom solutions for enterprise resource planning (ERP) system called SAP
- Created reports using **ABAP** and **SQL** for database queries and updates, order automation and GUIs
- Programmed cross-platform web applications using **JavaScript**, **XML** and **SAPUI5** (HTML5 framework)

Mechatronics Technologist | NRC Canmet Materials May 2022 - Aug 2022

- 3D modelled designs** and **simulations** to build corrosion and material testing **automation** projects
- Designed and **3D printed** a prototype of a sealing chamber with a **camshaft** and **servo** mechanism
- Programmed waypoints for a **robotic arm** to complete **automated** cycles of metal alloy casting

Research Assistant | McMaster University May 2021 - Aug 2021

- Developed and tested **Python** code to design hexagonal tiling of a prototype helmet for **optimal energy** absorption to study the impact of deformations on the helmet's ability to absorb impacts
- Named as a **coauthor** for the research presented at the CMSC conference

Education

Bachelor of Engineering, Mechatronics Co-op | McMaster University Sep 2020 - Apr 2025

- Cumulative GPA: 3.7
- Awards: President's Entrance Scholarship, Dundas Scholarship, McMaster's Summer Research Award
- Applicable Courses: Software Development, Operating Systems, Embedded Systems Design

Extracurriculars

Computer Chapter Vice Chair | McMaster IEEE Student Branch Jan 2022 - Apr 2023

- Led a sub team of 15 students to plan an intro to **Raspberry Pi** workshop involving two **Python** projects: a jukebox and an ADC visual display, to teach topics such as **GPIO**, **Audio**, **ADC**, and bit shifting
- Led the sub team project of a **Raspberry Pi controlled car** with **lane detection** using **Python OpenCV** and **brushed DC motors** controlled by a connected **Arduino** in **C**

Projects

Pacemaker Project Nov 2022

- Programmed the **state flow** of charge to capacitors for different modes of pacing in **Simulink**
- Created a **device controller-monitor** in **Python Tkinter** to send inputs through **serial communication**

Reaction Time Tester Embedded System Feb 2022

- Created a **finite state machine** to implement desired behaviour in a **memory efficient** way
- Coded in **C** to configure **GPIO** pins for **external buttons** to use **interrupts** to sense button pushes
- Used the **real-time clock** of the **STM32 microcontroller** to time reactions and display them on the **LCD**
- Recorded the fastest times to an external **EEPROM chip** with **I2C serial communication**

7-Segment Display Design Project Dec 2021

- Made a **7-segment display** cycle through my student number using **sequential logic design (K-map)** to make a decoder chip drive the display, taking inputs from **flip flops** to cycle states
- Created a digital prototype in **multisim** and built a working physical circuit with **logic gates**

Fly A Rocket! Program, Sensor Experiments Team | ESA Sep 2020 - Apr 2021

- Represented Canada as one of 24 students selected by the European Space Agency Education Office
- Completed course on necessary theoretical background and **Python** code used in the launch campaign

Skills

Software and Programming: Python, C, C++, JavaScript, SQL, XML, SAPUI5 (HTML5 framework), AWS, ABAP, Linux, Assembly, Matlab, Verilog, CAD (Autodesk Inventor, Solidworks), Simulink, Maple, Multisim

Hardware: STM32, Raspberry Pi, Arduino, Arm, 3D Printing, Oscilloscope, and Soldering